

AIMO Proof Pilot — Final Grade

Problem (#_ID): 2_ALTORO

Submission: model_7

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (docs/markschemes.pdf).

1. Final mark

Final mark (choose one):

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. **ME1**) where possible.

A complete, correct coordinate solution. It sets up coordinates, solves for H (with justification) and for D, E, F and Ω , justifies the harmonic relation $(B, C; D, X) = -1$, and computes X , Q , the midpoint P of HQ , and the lengths PQ and PE exactly (an arduous but correct calculation), finishing with the right answer. CG proposed reducing this to 6 over the sign handling in the derivation of X (lines 85–97); however, there is enough explanation around obtaining the coordinates of X for the step to stand, and the value $X = (170/3, 0)$ is correct. No minor error is warranted, so it scores **7**.